

1.13. PWA & Sync

- IndexedDB for local document and metadata cache
- Sync engine to:
 - Check for updates on re-connection
 - Upload pending changes
 - Resolve conflicts with version control

5. Scalability & Performance

- Target <1s search response for datasets with up to 10,000 documents.
- Lightweight client (under 5MB payload for initial offline sync).
- Indexed offline search limited to latest 500-1000 documents if resource constrained.

6. Compliance & Standards

- Must support SDS formats as per Australian regulatory standards (e.g., GHS classification).
- All metadata fields must map to industry-standard fields (Product name, CAS number, etc.).

7. Future Enhancements (Optional / Ideal) – to be priced as add-ons

- Barcode/QR scanning to retrieve SDS – faster identification of chemicals from container packaging. Critical during exposure or audit events
- Integration with APVMA database for live validation.
- AI-driven similarity search or SDS summarisation – Can surface first-aid steps and hazards more quickly in emergency scenarios
- Inclusion of functionality to enable / integrate CropLife's resistance management strategies -> build out for agronomists to assist with application planning and workability of different chemicals.